

Applied Cyber Threat Analysis – Network Traffic Analysis Lab

- This lab was conducted within the Virginia Cyber Range Environment.
- Report based off assignment assigned by George Mason University - Department of Information Sciences and Technology College of Engineering and Computing
- Cheche Agada. (2024). Email Header Analysis Lab IT-462: Applied Cyber Threat Analysis, George Mason University

Introduction

Network traffic analysis is an important part of digital investigations. Network traffic may contain evidence of malicious activity or network device failure/malfunction. In this lab, you will investigate a packet capture (PCAP) file using GUI tools as well as command line tools.

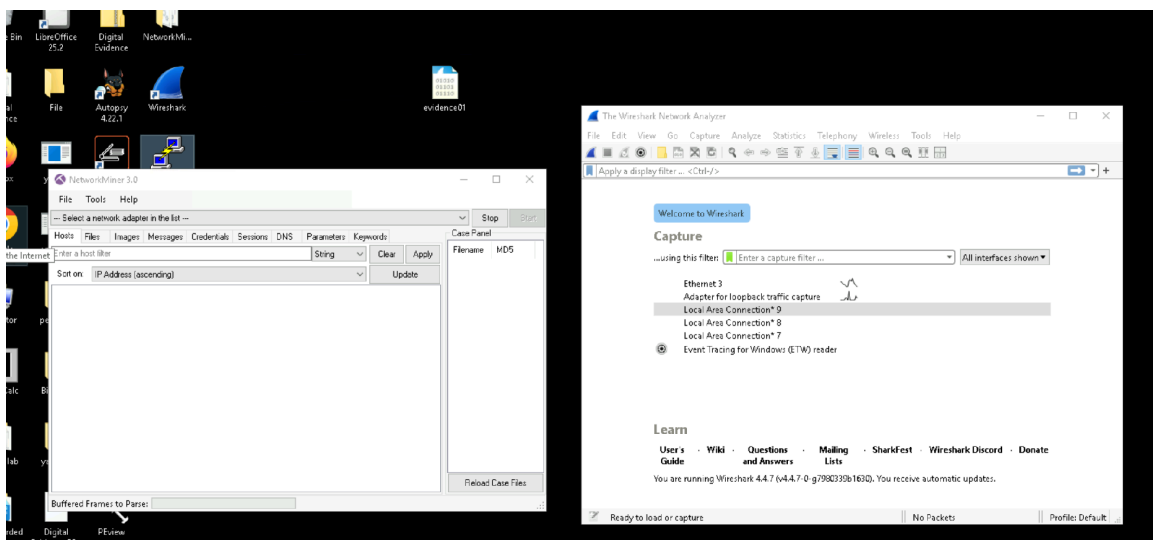
Objective

Conduct an analysis of a PCAP file.

Part 1

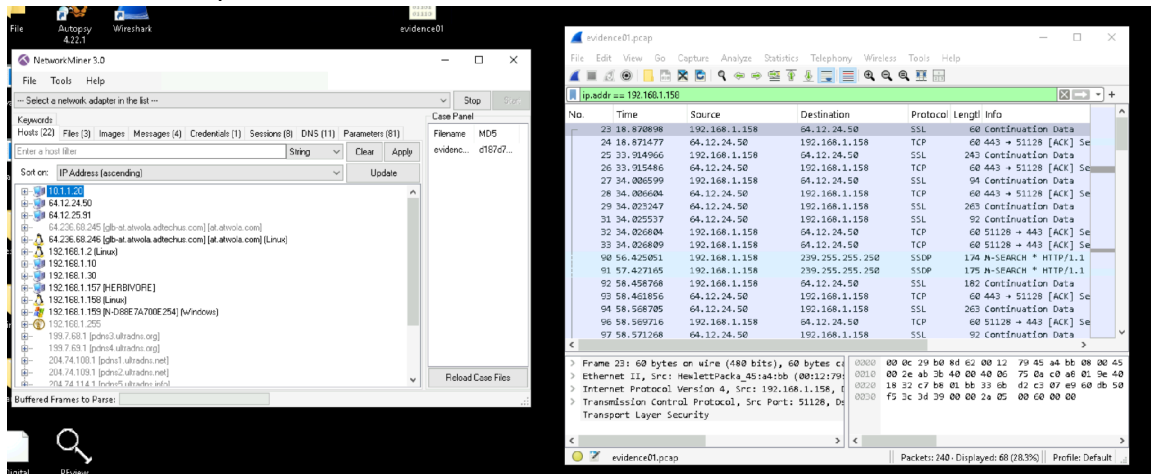
Task 1: Install Wireshark and NetworkMiner

- Download and install Wireshark (<https://www.wireshark.org/>)
- Download NetworkMiner (<https://www.netresec.com/?page=NetworkMiner>) – Both should have been installed in week 1.



Task 2: Analyze a Packet Capture (PCAP) file (GUI Tools)

- Download the file named 'evidence01.pcap' (Evidence File for Lecture) from this week's folder on Blackboard.
- Open the file in Wireshark as well as NetworkMiner.
- Use specific evidence (i.e. frame numbers, screenshots, etc.) from the PCAP file to answer the questions in exercise 1.

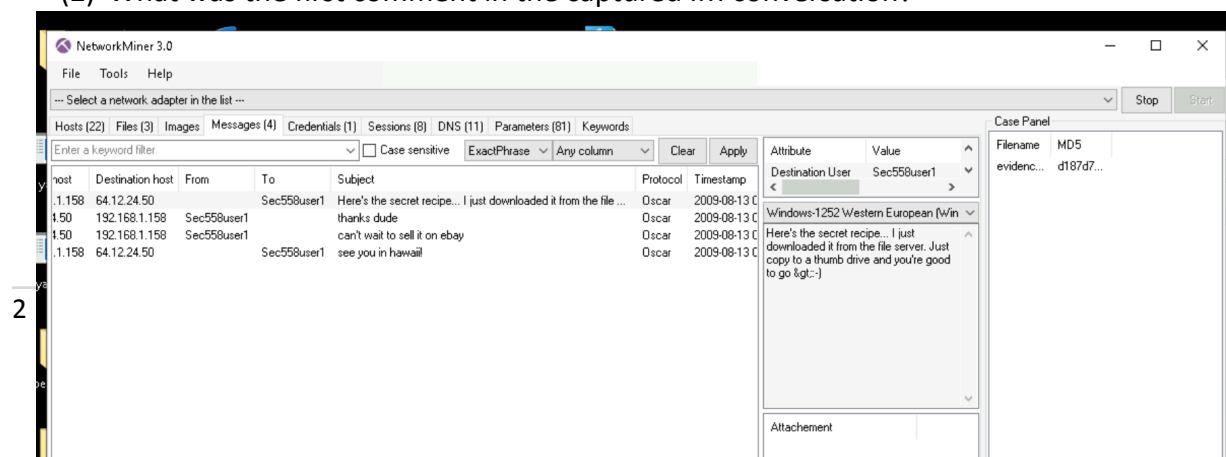


Exercise 1

Scenario: Anarchy-R-Us, Inc. suspects that one of their employees, Ann Dercover, is really a secret agent working for their competitor. Ann has access to the company's prized asset, the secret recipe. Security staff are worried that Ann may try to leak the company's secret recipe. Security staff have been monitoring Ann's activity for some time, but haven't found anything suspicious—until now. Today an unexpected laptop briefly appeared on the company wireless network. Staff hypothesize it may have been someone in the parking lot, because no strangers were seen in the building. Ann's computer, (192.168.1.158) sent Instant Messages (IMs) over the wireless network to this computer. The rogue laptop disappeared shortly thereafter. "We have a packet capture of the activity," said security staff, "but we can't figure out what's going on. Can you help?"

You are the forensic investigator. Your mission is to figure out who Ann was IM-ing, what she sent, and recover evidence including:

- (1) What is the name of Ann's IM buddy? **Sec558user1 – 64.12.24.50**
- (2) What was the first comment in the captured IM conversation?



“ Here’s the secret recipe... I just downloaded it from the file server. Just copy to a thumb drive and you’re good to go >-) ”

(3) What is the name of the file Ann transferred? **recipe.docx**

--- Select a network adapter in the list ---				
Hosts (22) Files (3) Images Messages (4) Credentials (1) Session				
Enter a keyword filter				
Frame nr.	Filename	Extension	Size	Source
112	recipe.docx	docx	12 008 B	192.168.1.1
230	size=120x90;noperf=1.html	html	375 B	64.230.1.1
233	size=120x90;noperf=1.js	js	335 B	64.230.1.1

(4) What is the magic number of the file you want to extract (first four bytes)?

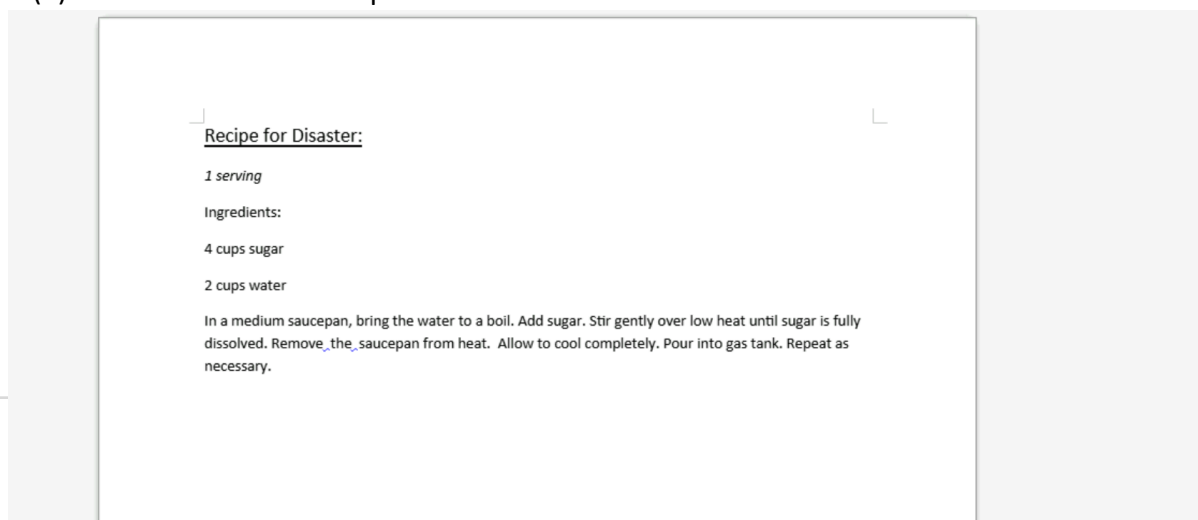
504B0304



(5) What is the MD5sum of the file? **8350582774E1d4dbe1d61d64c89e0ea1**

Name	recipe.docx
MD5	8350582774e1d4dbe1d61d64c89e0ea1
SHA1	11745854a7f8cd0c513dbaa695e84fde9fa0e581
SHA256	f0f74a982a814640aedaa5fd6542ac810e8c5e257552bcc024a5c8083
Path	C:\Users\student\AppData\Local\NetworkMiner\AssembledFiles\192
Size	12008

(6) What is the secret recipe?



Recipe for Disaster:

1 serving

Ingredients:

4 cups sugar

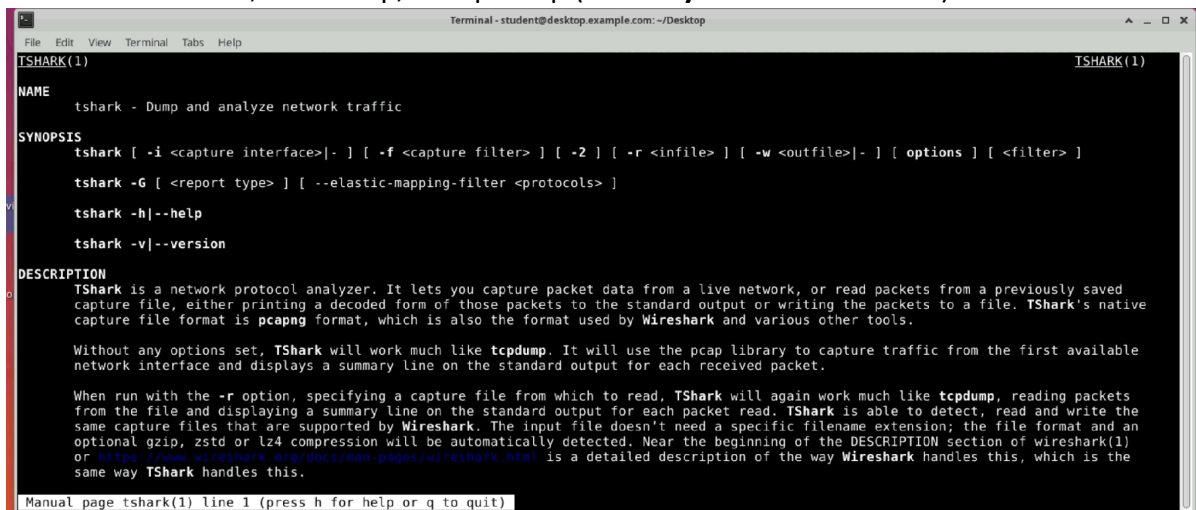
2 cups water

In a medium saucepan, bring the water to a boil. Add sugar. Stir gently over low heat until sugar is fully dissolved. Remove the saucepan from heat. Allow to cool completely. Pour into gas tank. Repeat as necessary.

Part 2

Task 1: Command Line Tools

- You may complete this portion of the lab using a Windows or Linux command line.
- Install TShark, WinDump, or Tcpdump (**You only need one of them**).



```
Terminal - student@desktop.example.com: ~/Desktop
TSHARK(1)
NAME
    tshark - Dump and analyze network traffic
SYNOPSIS
    tshark [ -i <capture interface>|- ] [ -f <capture filter> ] [ -2 ] [ -r <infile> ] [ -w <outfile>|- ] [ options ] [ <filter> ]
    tshark -G [ <report type> ] [ --elastic-mapping-filter <protocols> ]
    tshark -h|--help
    tshark -v|--version
DESCRIPTION
    TShark is a network protocol analyzer. It lets you capture packet data from a live network, or read packets from a previously saved
    capture file, either printing a decoded form of those packets to the standard output or writing the packets to a file. TShark's native
    capture file format is pcapng format, which is also the format used by Wireshark and various other tools.

    Without any options set, TShark will work much like tcpdump. It will use the pcap library to capture traffic from the first available
    network interface and displays a summary line on the standard output for each received packet.

    When run with the -r option, specifying a capture file from which to read, TShark will again work much like tcpdump, reading packets
    from the file and displaying a summary line on the standard output for each packet read. TShark is able to detect, read and write the
    same capture files that are supported by Wireshark. The input file doesn't need a specific filename extension; the file format and an
    optional gzip, zstd or lz4 compression will be automatically detected. Near the beginning of the DESCRIPTION section of wireshark(1)
    or https://www.wireshark.org/docs/man-pages/wireshark.html is a detailed description of the way Wireshark handles this, which is the
    same way TShark handles this.

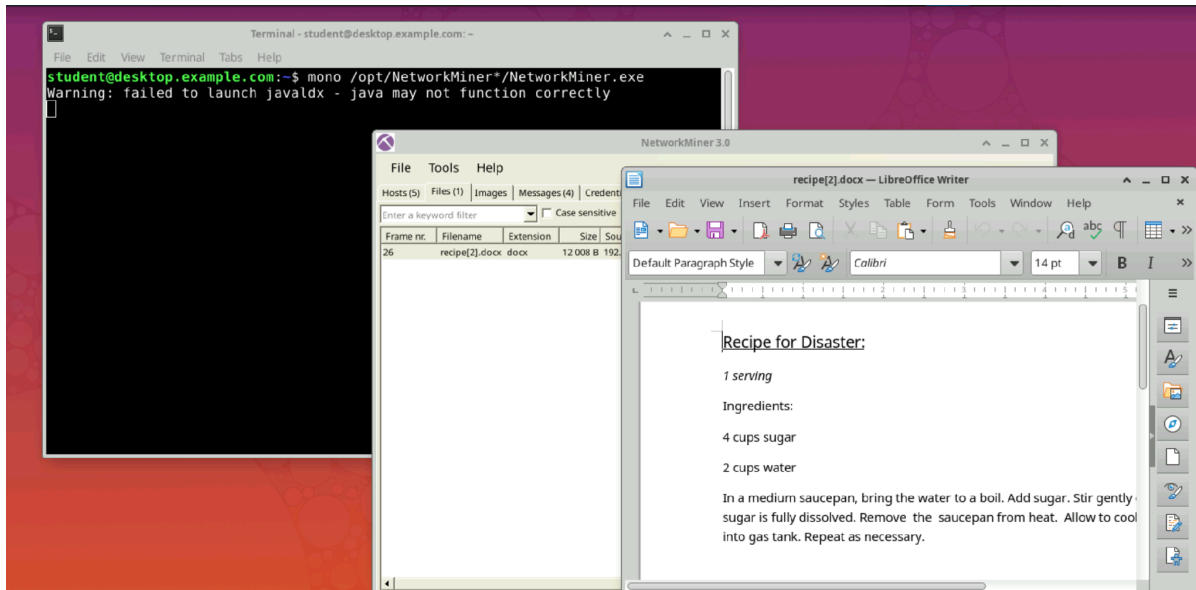
Manual page tshark(1) line 1 (press h for help or q to quit)
```

NOTE:

- **TShark** is the command line version of Wireshark and is installed with Wireshark. To run TShark on Windows, there are 2 options. You can navigate to the Wireshark folder on your command line and run TShark from there each time (not convenient), or you may add the Wireshark folder path to your system/environment variables so you can run TShark from any location on the command line (recommended).
- **WinDump** works on Windows while **Tcpdump** works on Linux.
- See the resource links below for help with installing these tools.

Task 2: Analyze a Packet Capture (PCAP) file (Command Line Tools)

(6) What is the secret recipe?



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